# Hypotheses

Building on research in collaborative HRI, three different hypotheses are formulated:

1. **H1:** Participants in condition C1 (test condition) will have lower cognitive load and better performance than condition C2 (remote environment).

2. **H2:** Participants in condition C1 (test condition) will have lower cognitive load and better performance than condition C3 (individual interface).

3. **H3:** Participants, who are spatially skilled or have gaming experience, will have less mental workload and have better performance results in all three conditions (display lower blink-rate elevations, smaller GSR spikes and shorter completion times and less collisions than low-skill users)

These hypotheses are motivated by prior research in HRI and human factors.

**H1** is supported by grounding in communication [1], which highlights the importance of copresence in reducing coordination costs and enabling low-effort collaboration. In C1, copresence allows operators to rely on implicit coordination through gaze, gesture, and shared attention. In C2, the loss of copresence forces coordination to become explicit via verbal communication, increasing coordination overhead and reducing situation awareness of the team member activity [2].

**H2** is grounded in findings [3] showing that shared collaborative interfaces support greater shared cognition and implicit coordination through continuous group awareness, whereas individual interfaces require explicit communication and increase coordination effort. In C3, operators must verbally reconstruct a shared mission picture before making joint decisions, while in C1 shared access to both vehicle feeds enables awareness of team activity without direct communication.

**H3** is based on evidence that spatial orientation skill and gaming experience improve efficiency in perspective-taking tasks critical for indirect-view teleoperation, leading to fewer collisions, lower NASA-TLX ratings, and faster task completion [4, 5, 6]. We therefore expect high-skill participants to show smaller GSR spikes, lower blink-rate increases, and shorter completion times across all three conditions compared to low-skill users.

**References**

**[1]** Clark, H. H., & Brennan, S. E. (1991). Grounding in communication.

**[2]** Riley, J. M., Strater, L. D., Sethumadhavan, A., Davis, F., Tharanathan, A., & Kokini, C. (2008, September). Performance and situation awareness effects in collaborative robot control with automation. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (Vol. 52, No. 4, pp. 242-246). Sage CA: Los Angeles, CA: SAGE Publications.

**[3]** Lowry, P. B., Roberts, T. L., & Romano Jr, N. C. (2013). What signal is your inspection team sending to each other? Using a shared collaborative interface to improve shared cognition and implicit coordination in error-detection teams. *International Journal of Human-Computer Studies*, *71*(4), 455-474.

**[4]** Hegarty, M., & Waller, D. (2004). A dissociation between mental rotation and perspective-taking spatial abilities. *Intelligence*, *32*(2), 175-191.

**[5]** Chen, J. Y., & Barnes, M. J. (2012). Supervisory control of multiple robots in dynamic tasking environments. *Ergonomics*, *55*(9), 1043-1058.

**[6]** Beatty, J. (1982). Task-evoked pupillary responses, processing load, and the structure of processing resources. *Psychological bulletin*, *91*(2), 276.